

Exp2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

PROGRAM:

```
clc;

clear;

q=input('Enter the value of q=');
r=input('Enter the value of r=');

efcilon=8.854*10^-12;

pi=3.14

E=q/4*pi*efcilon*r^2;

disp("The electric field is", [E]);

D=E*efcilon;

disp("The flux density is", [D]);

x = linspace(0, 10, 50);

y = E * exp(-0.1 * x);

z = D * exp(-0.1 * x);

figure();

subplot(2, 2, 1);

plot2d3(x,y);

xlabel("R");

ylabel("Electric field intensity");

title('Electric field intensity');

subplot(2, 2, 2);

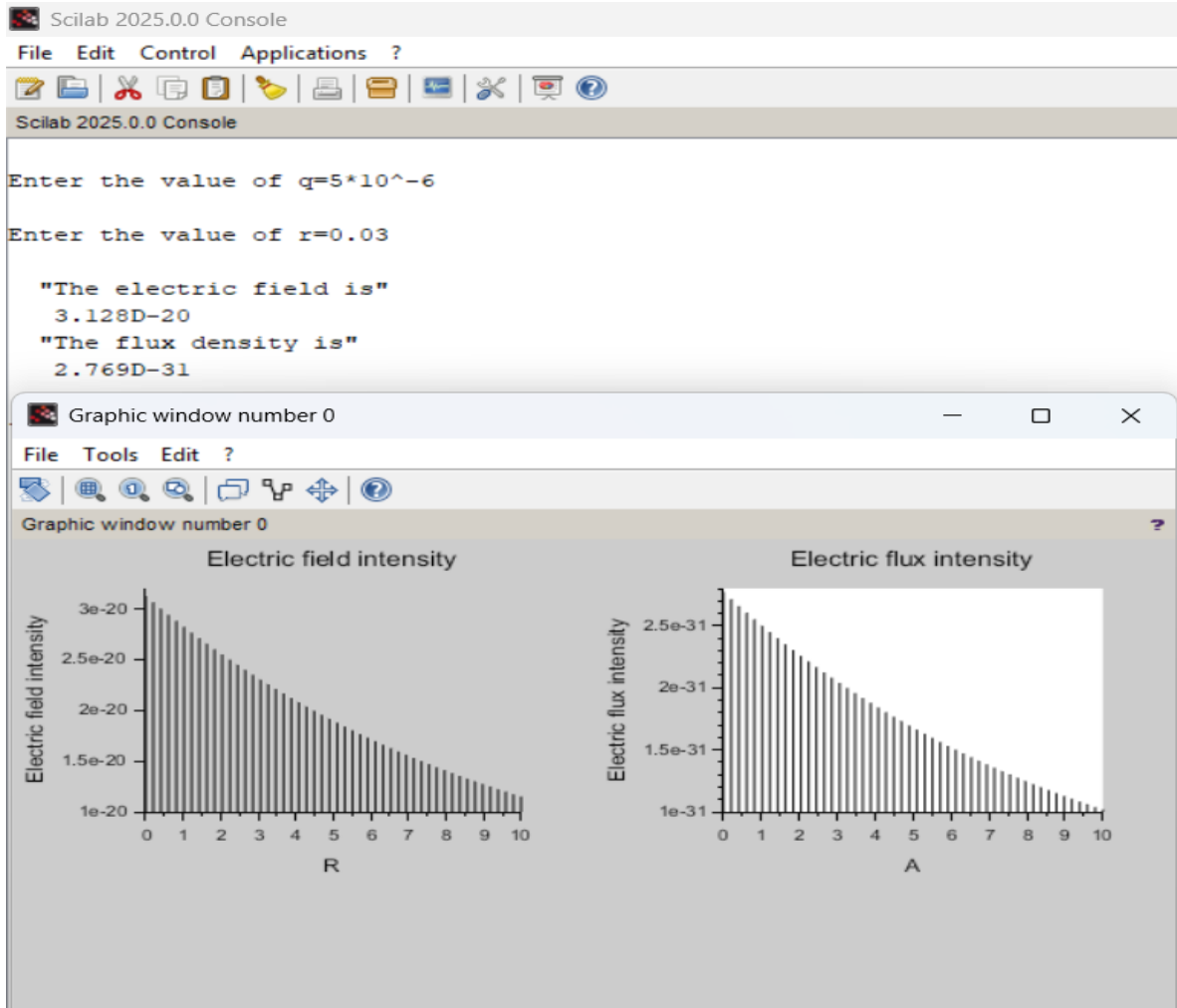
plot2d3(x,z);

xlabel("A");

ylabel("Electric flux intensity");

title('Electric flux intensity');
```

Output:



Calculation:

✓ $E \propto \frac{1}{r^2}$

$Q = 5 \times 10^{-6}$

$r = 3 \text{ cm} \Rightarrow 3 \times 10^{-2} \text{ m}$
 $= 0.03 \text{ m}$

$E = 8.854 \times 10^{12}$

$\Rightarrow$ Electric field

$\Rightarrow$ Electric flux Density (D)

$$E = \frac{Q}{4\pi r^2 \epsilon_0}$$

$$= \frac{5 \times 10^{-6}}{4\pi \times (0.03)^2} = \frac{5 \times 10^{-6}}{4\pi \times 9 \times 10^{-4}} = \frac{5 \times 10^{-6}}{1.130973 \times 10^{-3}}$$

$$E = \frac{5 \times 10^{-6}}{1.130973 \times 10^{-3}} = 3.128 \times 10^{-20}$$

$\Rightarrow$ Electric flux density

$$D = E \cdot \epsilon_0$$

$$= 3.128 \times 10^{-20} \times 8.854 \times 10^{12}$$

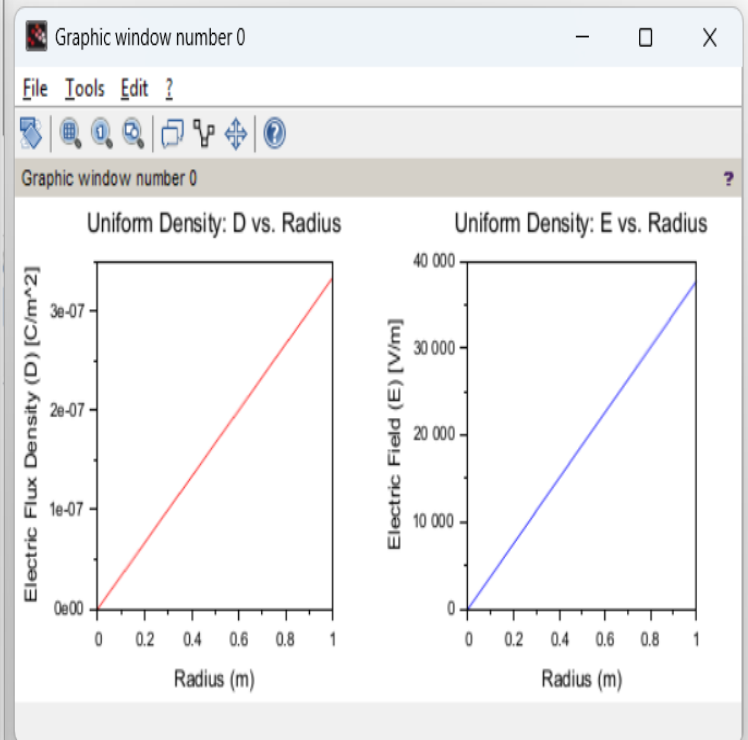
$$D = 2.769 \times 10^{-31}$$

0/p2

TEST CASE: 1

OUTPUT:

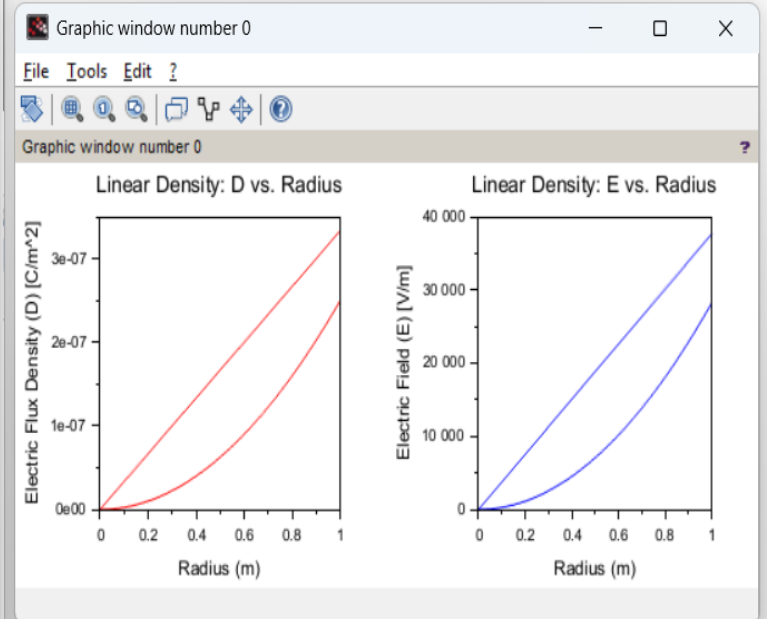
```
exp333.sci (C:\Users\pavan\Documents\beee\exp...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1 clc;
2 rho0 = 1e-6;
3 epsilon = 8.854e-12;
4 R = 1;
5 n = 100;
6 r = linspace(0, R, n);
7 D = zeros(1, n);
8 E = zeros(1, n);
9 for i = 1:n
10     ri = r(i);
11     Q_enclosed = rho0 * (4/3) * pi * ri^3;
12     if ri ~= 0
13         D(i) = Q_enclosed / (4 * pi * ri^2);
14     else
15         D(i) = 0; //Avoid division by zero
16     end
17     E(i) = D(i) / epsilon;
18 end
19 subplot(1, 2, 1);
20 plot(r, D, "r");
21 xlabel("Radius (m)");
22 ylabel("Electric Flux Density (D) [C/m^2]");
23 title("Uniform Density: D vs. Radius");
24 subplot(1, 2, 2);
25 plot(r, E, "b");
26 xlabel("Radius (m)");
27 ylabel("Electric Field (E) [V/m]");
28 title("Uniform Density: E vs. Radius");
29
```



TEST CASE: 2

OUTPUT:

```
exp333.sci (C:\Users\pavan\Documents\beee\exp...  — □ ×
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci ×
1 clc;
2 rho0 = 1e-6;
3 epsilon = 8.854e-12;
4 R = 1;
5 n = 100;
6 r = linspace(0, R, n);
7 D = zeros(1, n);
8 E = zeros(1, n);
9 for i = 1:n
10     ri = r(i);
11     Q_enclosed = rho0 * %pi * ri^4;
12     if ri ~= 0
13         D(i) = Q_enclosed / (4 * %pi * ri^2);
14     else
15         D(i) = 0;
16     end
17     E(i) = D(i) / epsilon;
18 end
19 subplot(1, 2, 1);
20 plot(r, D, "r");
21 xlabel("Radius (m)");
22 ylabel("Electric Flux Density (D) [C/m^2]");
23 title("Linear Density: D vs. Radius");
24 subplot(1, 2, 2);
25 plot(r, E, "b");
26 xlabel("Radius (m)");
27 ylabel("Electric Field (E) [V/m]");
28 title("Linear Density: E vs. Radius");
29
```



TEST CASE: 3

OUTPUT:

```
exp333.sci (C:\Users\pavan\Documents\beee\exp... - □ X
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1 clc;
2 rho0 = -3e-6;
3 epsilon = -8.854e-12;
4 R = 2;
5 n = 200;
6 r = linspace(0, 4, n);
7 D = zeros(1, n);
8 E = zeros(1, n);
9 Q_total = rho0 * (4 * pi * R^3 / 3);
10 for i = 1:n
11     ri = r(i);
12     if ri < R then
13         Q_enclosed = rho0 * (4 * pi * ri^3 / 3);
14     else
15         Q_enclosed = Q_total;
16     end
17     D(i) = Q_enclosed / (4 * pi * ri^2);
18     E(i) = D(i) / epsilon;
19 end
20 subplot(1, 2, 1);
21 plot(r, D, "r");
22 xlabel("Radius (m)");
23 ylabel("Electric-Flux-Density (D) - [C/m^2]");
24 title("Uniform-Charge-Density: -D-vs.-Radius");
25 subplot(1, 2, 2);
26 plot(r, E, "b");
27 xlabel("Radius (m)");
28 ylabel("Electric-Field (E) - [V/m]");
29 title("Uniform-Charge-Density: -E-vs.-Radius"); clc;
30 rho0 = -3e-6;
31 epsilon = -8.854e-12;
32 R = 2;
33 n = 200;
34 r = linspace(0, 4, n);
35 D = zeros(1, n);
```

